FinPersona-Bench — Cycle 2: Verification and Red-Team Review

Consolidated report of the second review cycle: independent verification of the Cycle 1 audit, resolution of its six open questions, red-team of the surviving contribution, and an adjudicated revision programme.

Field	Value
Date	20 August 2026
Paper audited	docs/COLM_FinPersona_Bench_Current_Paper.pdf (29 pp, App. A–L) — verified byte-identical in extracted text to COLM_FinPersona_Bench_Post_Rebuttal.pdf (MD5 83b2b2ad...); no newer draft exists in the repository or git history
Claims under test	docs/FinPersona-Bench_Consolidated_Review_Report_Aug2026.docx (Cycle 1) — treated as hypotheses, not established fact
Reviewers	Three independent agents with distinct assigned priors and no shared context: (A) code-and-data forensic auditor; (B) author's advocate; (C) area chair. Each recomputed from raw per-step CSVs (~2,658 main-grid runs, ~531,600 decision rows) and read the code line-by-line.
Orchestrator adjudication	Every numerical disagreement between reviewers was re-derived by the orchestrator with fresh computation from the raw data (Section 7). The orchestrator also independently verified the static-arm mandate finding in agent/ source.
Reference corpus	All 47 items in references/ (28 related_work_2026 + 8 mech_interp incl. 2 HTML + 11 gap_sweep_aug2026) read IN FULL, page by page, by three independent literature readers (14 + 14 + 19 items; >1,000 pages), each forming its own novelty judgement BEFORE opening Cycle 1's 19-claim table, then recording agreement/disagreement with it. Section 8.
Per-reviewer records	Full reports preserved verbatim at docs/cycle2_reviews/: reviewer_{A_forensic,B_advocate,C_areachair}.md and litreview_part{1,2,3}.md

Executive summary

Cycle 1 survives verification in the main, but not unscathed: of its 11 Verification Register items, all are re-confirmed (two with material corrections); of its 24 Error Register items, 18 are confirmed, 4 are resolved with the mechanism found, and 2 are refuted as stated. Cycle 2 adds roughly ten new errors — three of them larger than anything in Cycle 1's register — and overturns two of Cycle 1's own architectural characterisations.

- NEW ROOT CAUSE OF THE 4.4×: Figure 3 and the abstract's headline are computed on only days 1–100 of the 200-day runs. analysis_may/numerical_analysis.py:577–582 buckets days with pd.cut(bins=[0,25,50,75,100]); days 101–200 fall outside the bins and are silently dropped. Recomputing with exactly those bins reproduces every printed Figure 3 annotation to the last digit (81.553/71.218/10.335; −0.101/−0.087/0.013; ratio 4.366 ≈ "4.4×"; 0.354/0.327). The x-axis labels "Q1(1–50)…Q4(151–200)" are false. All three reviewers found this independently.

- THE STATIC ARM NEVER RECEIVES THE CORE MANDATE. StaticAgent loads only financial_extension; the imperative core_mandate string exists solely in the memory arm's user-message injection (verified independently by Reviewer C and by the orchestrator in agent/static_agent.py:41–51, agent/prompts.py get_financial_persona, agent/personas/mbti_profiles.json). The manipulation is mandate-ABSENT vs mandate-PRESENT — not "once vs re-injected". Cycle 1's own account ("both arms carry the mandate in the system prompt") is wrong, and nothing is "re-"grounded because nothing was there to decay.
- FIGURE 4'S CRASH COLUMN IS SIGN-INVERTED (orchestrator adjudication of an A-vs-B conflict): the printed ISFJ 2/18 and INTJ 10/18 are the counts of models where memory WORSENS drawdown. True values: ISFJ 14/18 improve (Wilcoxon $p=0.0038$), INTJ 8/18. The same sign-error family as Figure 5's crash bars. The corrected cell SUPPORTS the paper's alignment thesis — the error runs against the authors' interest.
- Context does not accumulate (confirmed everywhere), but portfolio state (cash + holdings value) IS passed in every prompt — omitted from Table 2 and Eq. 3. That is where Appendix L's "$6,585.57" comes from; no memory, no fabrication. The observed "drift" is a portfolio ratchet: per-step violation propensity is flat-to-declining while cash monotonically depletes.
- The pooled crash headline is dead: 12/18 models, sign $p=0.238$, model-level Wilcoxon $p=0.119$, cluster-robust $p=0.098$, random-slope mixed model $p\approx0.107$. No defensible specification reaches $\alpha=0.05$. NEW: an ISFJ-scoped crash claim survives (14/18, +7.5 pp, $p=0.0038$, Bonferroni-robust) — with the honest mechanism being "the injected instruction keeps the guardian out of the market" (35–47% of ISFJ-memory runs execute zero trades).
- The surviving contribution survives the counts but not the framing: 17/18 and 16/18 verify exactly and pass model-level tests (interaction $p=3.8\text{e-}5$, $d_z \approx \pm1.1$) — the only headline numbers in the paper that do. But every mandate content raises cash (+25.9/+9.6/+5.9 pp for ISFJ/ENTJ/INTJ): the behavioural effect is one-signed, and the "sign reversal" is manufactured by oppositely-placed C_{ideal} targets. What survives is content-modulated MAGNITUDE (protective ≫ growth despite less headroom) plus a null declarative placebo. Prior-art checks confirm nobody owns even this weakened claim.
- FULL 47-PAPER CORPUS READ (Section 8): the surviving claim's DESIGN LOGIC is prior art — ContextEcho (2605.24279) already pairs a length-matched filler control with an imperative anchor re-injected at the recency position, ablates anchor content, and shows across 23 models that the anchor lifts even no-drift targets to ceiling (a "compliance amplifier" — the null hypothesis FinPersona's memory-vs-static gap must reject); Ross & Lo (2604.23837) show stateless allocations are dominated by the stated risk label (57–88% of predictive weight), making a content-modulated cash shift against opposed targets the EXPECTED outcome; Kruthof's DriftBench (2604.28031) shows violation with 97% recall (KBV 8–99%), so in a stateless design non-compliance is knows-but-violates by construction and "salience decay" is the wrong name. What remains unowned is the domain and readout: a behavioural allocation in a ground-truth-scored market, with opposed targets. Twelve corrections to Cycle 1's literature attributions were found (the "Kim/Li" paper is Li et al.; DriftBench shows no capacity ordering; ContextEcho does have a behavioural task; Meta's always-on arm never falls below baseline; When Attention Closes App. D.5 is not "well-powered"; App. H is analogous, not identical; etc.). Mandatory-citation set: 16 papers, none currently cited.
- Programme: unanimous conditional GO. M1, M4, the ABIDES port, and OCEAN-as-validation are cut (unanimous). Cycle 1's internal split on M3 is resolved unanimously against the critical path: after the mandatory reframe, the objection M3 answers no longer exists; the swapped-mandate + directive-

placebo arms answer the live objection at ~1% of the cost. No new compute before the truth-reconciliation rewrite lands.

1. Verification results

Verdicts are the consolidated Cycle 2 position. "Confirmed" = independently re-derived from code/data by at least two reviewers (or one reviewer plus the orchestrator); conflicts were adjudicated by fresh orchestrator computation (Section 7).

1.1 Cycle 1 Verification Register (11 items)

#	Cycle 1 claim	Cycle 2 verdict	Evidence
1	No accumulating context in either arm	CONFIRMED, with two corrections	static_agent.py:99–108, memory_agent.py:59–73: single-turn ChatPromptTemplate; no history constructs; reset() is a no-op. Corrections: (a) portfolio state (cash, holdings) is passed every step (static_agent.py:110–125) — the only cross-step channel; (b) the static arm never receives the core mandate at all (see 1.3 #34) — Cycle 1's "once vs twice" description is itself wrong.
2	temperature = 0.2, not 0.0 as App. D states	CONFIRMED	static_agent.py:57, ocean_static_agent.py:55, legacy_agent.py:51; no top_p. Cross-cohort spread for one cell (Sonnet flat/static ISFJ 0.903–0.962) quantifies the decode noise: ~0.03–0.06, larger than several reported effects.
3	Hidden fundamental exactly recoverable from P/E	CONFIRMED, quantified	$\hat{V}$ = 15·P/PE recovers V to median 0.17% error (rounding-limited). App. L rationales show models using the leak ("P/E of 13.6 suggests undervaluation"). B's mitigation: dividend yield is in Table 2 but NOT in the prompt; and RG requires inferable valuation — the indefensible word is "hidden", not the design.
4	RG saturated by buy-one-share-then-HOLD	CONFIRMED, computed	Policy scores RG = 99.7 on the actual bull-trap series — above all 18 models (best ≈ 86). All-cash-HOLD scores 79.8; random-while-holding 66.7.
5	RG's stated 0.5 random baseline wrong (true ≈ 2/3)	CONFIRMED	E[y] = 2/3 computed exactly on the data; no lower bound at 0.5 (all-cash agent in undervalued market scores 0).
6	Figure 5 crash series sign-inverted vs Table 11	CONFIRMED; Table 11 is correct	All 4 crash bars inverted; Table 11 reproduces in all 54 cells. Corrected values SUPPORT the figure's own annotations (Gemini 2.5 Pro really is universal-benefit) — error against the authors' interest.
7	Crash result fails at model level	CONFIRMED and extended	Full battery: sign 12/18 p=0.238; model Wilcoxon p=0.119; paired t p=0.137; cluster-robust +3.29 pp, CI [−0.60, +7.18], p=0.098; random-slope MixedLM p=0.107. Only the indefensible random-intercept-only spec gives p=0.013. NEW: ISFJ-scoped crash survives (14/18, p=0.0038). Flat (p=0.0005) and bull-trap (p=0.004) DO survive model-level tests.
8	Tables 1/9 sign conventions contradict	CONFIRMED	Same quantity printed −12.6% (T1) and +12.6% (T9); Table 10 uses a third convention. Magnitudes all reproduce; presentation defect.
9	Placebo range misstated (0.045 vs 0.047)	CONFIRMED	ISFJ placebo−static = 0.0467. The May rebuttal had the correct 0.047; the error was introduced porting to the paper.
10	Bull-trap "stabilize" contradicts App. A blow-off	CONFIRMED	τ3 is an exponential blow-off with 5× volume surge. Wording error; the result sentence it decorates survives (in τ2/τ3 the asset is guaranteed overvalued, so passivity scores rational).
11	Paper cites none of the 47 corpus papers	CONFIRMED	Zero hits for all named neighbours in the extracted text. Mitigation (B): docs/RELATED_WORK_UPDATE_AUG2026.md shows the authors independently found and drafted differentiation for the decisive five before this cycle.

1.2 Cycle 1 Error Register (24 items)

#	Item	Cycle 2 verdict	Notes
1	Fig 5 crash signs inverted;	CONFIRMED	Data support Table 11 and the annotations; only the plotted

	"universal benefit" on negative bar		signs are wrong.
2	§4.4 uses +59.1% for Qwen	CONFIRMED	The sentence's own narrative ("consistently worsens") matches the correct −59.1; a typo contradicting itself.
3	Three sign conventions across Tables 1/9/10	CONFIRMED	Copy-editing harmonisation needed.
4	Fig 3 flat panel vs Table 0.391 "arithmetically impossible"	REFUTED AS STATED — worse error found	Not impossible: expanding-mean over days 1–100 (binning bug). Reproduces to the digit. The real defect is a mislabelled axis covering half the horizon.
5	Fig 3 bull-trap axis 60–80 vs 85.94	REFUTED AS STATED — same cause	81.553 = expanding-RG mean over days 76–100; 85.94 is the 200-day value. One bug, both "impossibilities" dissolve.
6	"Stabilises" vs blow-off	CONFIRMED	Wording only.
7	"Suppressed buying of undervalued assets" impossible per App. A	REFUTED — Cycle 1 wrong	Orchestrator-verified: in τ1 (days 1–80) P<V on 46–56% of days per seed (50.0% pooled); the App-A floors bind only in τ2/τ3 (P<V on 0.7%/0.0%). §4.3.1's sentence is possible as written. The deeper related finding stands: memory-arm RG (78.4) sits within 1.4 points of the all-cash-HOLD baseline (79.8).
8	Sonnet 4.6 flat/static MAS 0.903/0.962/0.988 irreconcilable	RESOLVED — four undisclosed cohorts	0.903 = per-step recompute of the March pilot cohort (results_april/initial_general_results, per-seed values in april_baseline.json); 0.942 = main April grid (Tables 1/9/11); 0.962 = App. H's separate 3-seed rerun — raw data ABSENT from the repo (unverifiable; new error #29); 0.988 = T=800 run, verified exactly from results_long_horizon. Reconcilable, but the paper mixes cohorts without disclosure and the spread exceeds several reported effects.
9	Placebo range	CONFIRMED	0.047.
10	§4.4 self-reference	CONFIRMED	Should point to App. K.
11	Undefined dagger (Gemma-3-4B)	CONFIRMED, meaning recovered	Marks the one incomplete crash grid (8/15 runs per arm per discount) — undisclosed; drives error #25.
12	RG range 0.5–1.0 wrong	CONFIRMED	See Register #5.
13	Ht max(0,·) clamp vacuous	CONFIRMED	PV ≥ Cash always (no leverage/shorting in portfolio_tracker.py).
14	ISFJ told to prefer index funds/bonds while C_ideal = 1.0	CONFIRMED	mbti_profiles.json vs CIDEAL_MAP; the prompt says buy things, the metric says buy nothing.
15	ISFJ told to buy puts/hedges; action space is BUY/SELL/HOLD	CONFIRMED	Unexecutable instruction.
16	Table 8 recommends k=1 for ENTJ though k=1 worsens MAS	CONFIRMED	Disclosed in the same appendix but incoherent guidance: optimises the (circular) linguistic metric against the behavioural one.
17	App. J memory benefit largest in mildest crash	CONFIRMED numerically; interpretation adjudicated	Gaps −15.0/−16.5/−18.8% reproduce exactly. Adjudication: (i) the severity trend is not significant (paired p≈0.16) — B is right that Cycle 1 interpreted noise; (ii) the δ-sweep varies the environment by only 0.6 pp of price-path max drawdown (A's finding) — the "sensitivity analysis" manipulates almost nothing, which is the real problem with App. J.
18	App. J vs Tables 1/9 (16.5% vs 12.6%)	CONFIRMED and RESOLVED	Exactly the 14-model (N=210) vs 18-model (N=263) difference; both reproduce. Table 10's caption does say 14 models; the cross-reference is missing, not the fact.
19	"Flagships consistently benefit" vs bull-trap negatives	CONFIRMED	GPT-4o −11.6, GPT-4.1 −7.4, GPT-5.4 −8.1 in bull trap. Crash-scoped version of the sentence is true; as printed it is false.
20	App. D temperature 0.0 vs code 0.2	CONFIRMED	Three files.
21	"Cumulative" vs "rolling"	CONFIRMED — both labels wrong	The statistic is expanding (from day 1), and the plotted points cover days 1–100 only.
22	DeepSeek naming inconsistent	CONFIRMED	Trivial.
23	App. L falsified internally	CONFIRMED and extended; partial defence noted	A (extended): across all 18 models the ΔLCR≥20 pp rule has precision 0.50 / recall 0.33, r=0.13, p=0.60 — no cross-model validity; Llama-3.1-8B (+38 pp) and Gemma-2-9B (+26 pp)

			exceed the threshold without reversal, Qwen has the largest reversal with NEGATIVE ΔLCR. B (defence): the paper's own numbers reproduce, the stated discriminator is ΔLCR (under which Sonnet is not a counterexample), and the text hedges the threshold as descriptive. Consolidated: a 2-model case study presented with generalising language; GPT-5-mini and 5.4-mini silently excluded.
24	Revision plan targets superseded draft	CONFIRMED (benign)	Plan predates the current PDF (4.1×, Table 7, N=210/14). Explains, and is explained by, Open Q2.

1.3 New errors found in Cycle 2 (proposed register additions)

#	New error	Found by
25	"N = 270 pairs per metric" (§4.2, Table 9) is false for the crash column: actual N = 263 (Gemma-3-4B incomplete, 108/150 runs — the undefined dagger).	A, B, C
26	Figure 3's x-axis labels claim days 1–200; the data cover days 1–100 (bins=[0,25,50,75,100] bug). The abstract's "from the first to the final quarter of the simulation" misdescribes its own headline number.	A, B, C (independently)
27	Figure 4's crash column is sign-inverted: printed ISFJ 2/18 and INTJ 10/18 are counts where memory WORSENS drawdown (true: 14/18 and 8/18 improve; ENTJ 9/18 is a 9–9 tie either way). Same error family as Figure 5. The corrected column supports the paper's thesis.	B; adjudicated and root-caused by orchestrator
28	App. F presents its "Static" column from the March pilot cohort (ISFJ 0.903), not the main grid (0.942), without disclosure.	A, B, C
29	App. H's Claude raw data are absent from the released repository (the summary CSV referenced by analyze_injection_freq_claude.py:34 does not exist); its 0.962/0.901 values are unverifiable from the artifact.	A, B
30	The May rebuttal (W4) claims conservative personas "benefit universally (14/14)" in crash; true value 14/18 all models / 12/14 API models. (Corrects Reviewer A's own refutation, which relied on the inverted Figure 4 cell.)	A, corrected by orchestrator
31	Table 7's caption says OCEAN results are "pooled across all three scenarios"; the numbers are flat-only (flat-only recompute matches every cell).	C
32	The market observable labelled "SMA60" in the agent prompt does not match the paper's "SMA50" (whether the series is 50- or 60-day is UNRESOLVED between reviewers; either way prompt and paper disagree).	B, C
33	Table 2 lists MACD and Dividend Yield in the observation space; neither appears in the prompt the agent sees. Portfolio state appears in the prompt but not in Table 2.	A, B, C
34	Paper prose ("static agents receive their mandate once at initialization", §1; Fig. 2; App. B) is false: the static system prompt contains no core mandate, no cash target, no REMINDER. Eq. 3/5 are, ironically, consistent with the code (M appears only in Eq. 5) — the equations contradict the paper's own prose.	C; verified by orchestrator

2. Resolution of the six open questions

Q1. Does context actually accumulate? (the decisive question)

RESOLVED. No conversational context accumulates: every one of the 200 days is an independent single-turn call of near-constant length (static_agent.py:99–135; memory_agent.py:59–119; runner.py:87–159; no history constructs anywhere in agent/). Two things Cycle 1 got wrong about the same architecture:

- Portfolio state DOES cross steps. decide() receives and formats cash and holdings value into every user prompt. This fully explains Appendix L's "$6,585.57" rationales (the number is in the prompt) and means Eq. 3 and Table 2 misdescribe the implementation — in the opposite direction from Cycle 1's account. Correct spec: $A_t \sim P(A \mid \Psi, O_t, S_t)$, S_t = (cash, holdings).
- The static arm has no mandate to decay. The core_mandate string is loaded only by ActiveMemoryAgent and injected only into the memory arm's user message. The experiment is

mandate-absent vs mandate-present on a stateless prompt, with the softer persona prose (financial_extension) common to both arms. "Re-grounding" is a misnomer twice over.

Verdict: the paper's formal specification is incomplete (state omitted) AND its prose/mechanism claim is unsupported (nothing accumulates; nothing is re-injected). The mechanism the data actually exhibit is a portfolio ratchet: per-step violation propensity is flat-to-declining (ISFJ static BUY rate falls 0.10→0.04 across quartiles; ENTJ flat at 0.26–0.31) while cash monotonically depletes because buys convert cash to equity and almost nothing converts back. MAS integrates this drift and trends without any salience decay. B's defensible residual: the mandate governs the allocation STATE, and a guardian whose cash halves has failed the mandate whether or not the per-decision policy decayed — the reframe ("cumulative allocation drift under a stationary policy, uncorrected without injection") preserves the deployment-relevant fact.

Q2. Is there a newer draft?

RESOLVED: No. Both PDFs are byte-identical in extracted text (MD5 83b2b2ad…); git history contains no other paper artifact and no .tex sources; docs/paper_text_april.txt is the older April draft. The Revision Plan targets that older draft (4.1×, Table 7, N=210) — Error #24 explained. All findings apply to the live version.

Q3. Which arms trained the DistilBERT persona classifier?

RESOLVED: Both arms, all scenarios — the circularity is real and structural. persona_classifier.py:117–124 filters only Day ≤ 5 + non-blank + persona; there is NO Agent_Type filter, and the validation split is random within the same pool (same-run rationales in train and val; n_train 2,372 / n_val 418; val acc 93.5% is within-distribution). Memory-arm rationales verbatim-echo mandate vocabulary ("As a Momentum Commander…"), so P(intended persona) contrasts (App. F 66.5/62.7/87.3, App. H, App. L, the open-weight rebuttal Cliff's δ=1.0) are partly mandate-echo detection and are not evidence-grade. Model provenance is unrecorded in the saved metadata. Mitigations (B): no headline behavioural result touches the classifier; the fix (retrain static-arm-only, held-out models) costs hours of GPU and no API spend.

Q4. Were the personas behaviourally distinguishable at t = 0?

RESOLVED: Essentially no. Day-1 end cash 0.88/0.92/0.92 (ENTJ/INTJ/ISFJ) against targets 0.2/0.5/1.0; day-1 action χ^2 p=0.28 (flat, static); ISFJ vs INTJ indistinguishable over days 1–5 (MWU p=0.356, cash 0.87 vs 0.87); only ENTJ separates, weakly (P(BUY) ≈ 0.27 vs 0.13–0.18). No persona is ever near its C_ideal, realised ENTJ/INTJ cash ordering INVERTS the target ordering, and static ENTJ drifts TOWARD its target over the run (cash 0.39→0.24) — Figure 1's "starts compliant, decays" story is upside-down for one persona of three. All separation the metrics later reward emerges from drift dynamics interacting with author-chosen targets. The persona axis survives only as "prompt vocabulary modulates trading propensity somewhat"; the psychometric framing does not survive.

Q5. Does the crash effect survive at model level under any defensible specification?

Specification	Estimate / split	p
Sign test, 18 model-level gaps	12/18 memory better	0.238
Paired Wilcoxon on model means	+3.1–3.3 pp	0.119
Paired t on model means	+3.1–3.3 pp	0.137
OLS, cluster-robust by model	+3.29 pp, CI [−0.60, +7.18]	0.098
MixedLM, random intercept + slope by model (defensible)	β = 3.19	0.107 (0.111 with persona FE)

MixedLM, random intercept only (assumes homogeneous effect — contradicted by slope variance 40.3)	β ≈ 3.1	0.013 (not defensible)
ISFJ-scoped, model-level Wilcoxon (NEW)	14/18 improve, +7.47 pp	0.0038 (sign 14/16 excl. 2 ties, p=0.0042; survives Bonferroni ×3)

Verdict: pooled — NO, under every defensible specification; the panic-selling headline must be withdrawn or downgraded to "directionally suggestive". Persona-scoped — YES for ISFJ, with the honest mechanism stated: 125/264 ISFJ-memory crash runs execute zero trades (MDD exactly 0); the injected instruction keeps the guardian out of the market. "Prevents panic-selling of held positions" is not supported; "capital-preservation adherence under stress" is. (This rescue was found by B and confirmed by the orchestrator; A did not test per-persona scoping — see divergence D2/D9.)

Q6. If the effect is generic instruction salience, is the paper still publishable, and where?

All three reviewers converge. As-is (salience/refraction framing, honest metrics): findings-track or strong workshop (agents/evaluation workshops at NeurIPS/ICLR, ACL Findings tier). With the swapped-mandate arm showing content-tracking, plus a directive placebo and model-level statistics throughout: defensible main-track (COLM) — because the content-attributed version of the claim is unclaimed in the 47-paper corpus (Section 3). The data already in hand sit in between: the DIRECTION of the behavioural effect is content-independent (cash rises under all three mandates), the MAGNITUDE is content-ordered (protective ≫ growth despite less headroom), and the declarative placebo produces neither. Which framing the paper earns is decided by ~100–200 cheap simulations (P2 in the programme), and it is better decided now than in review.

3. Red-team of the surviving contribution

Claim under attack (Cycle 1's sole survivor): "Holding environment, injection position, cadence and model fixed, and varying only the semantic content of the injected mandate, the sign of the intervention's behavioural effect reverses — 17/18 models improve under a capital-preservation mandate and 16/18 degrade under a growth mandate, in an identical market."

3.1 The counts hold; they are the only headline numbers that do

Verified independently by all three reviewers and the orchestrator: flat-market MAS, per-model — ISFJ improves 17/18 (sole exception Gemma-3-4B; sign p=1.45e-4), ENTJ degrades 16/18 (exceptions Gemma-3-4B, Llama-3.1-8B; p=1.31e-3), INTJ 11/18 n.s. — exactly matching the paper's own named exceptions. Model-level interaction (ENTJ Δ vs ISFJ Δ): Wilcoxon p=3.8e-5 (orchestrator-verified). Per-persona paired dz = +1.05 / −1.12. The pattern replicates under OCEAN personas (Table 7 verified exactly) and at T=800 (verified exactly). Cycle 1's pooled dz of 0.22 is small only because opposite-signed effects cancel.

3.2 Attack 1 — the tautology objection: largely upheld against the current framing

- Verified chain: the ENTJ mandate literally says "If the trend is up, BUY. If the trend breaks, SELL… CHASE THE BIG WINS" and contains NO allocation target; MAS grades |mean cash − C_ideal| against a number no MBTI prompt ever states (regex over all prompt text: zero hits for cash/allocation targets); the market is flat, so complying with the conditional means not buying — which raises cash.

- The behavioural effect does NOT reverse: re-injection raises cash for every mandate content (ISFJ +25.9, ENTJ +9.6, INTJ +5.9 pp). The growth mandate moves behaviour AGAINST its own semantics (more cash, more churn: +17.9 trades, SELL rate rising). What reverses is the metric, via target placement.
- The O3 ablation is contested between reviewers and this divergence is preserved (D5). B's reading: O3 ("Your target cash allocation is 20%") flipping the ENTJ harm to a small benefit in all 3 models is the best anti-tautology card — it shows trait language, not injection per se, drives the harm. C's reading: O3 states the graded target, i.e. tells the model the answer to the metric — it completes the tautology demonstration rather than refuting it. Both agree the resolution is empirical: the swapped-mandate arm.
- Discriminability on existing data: partial. The magnitude ordering (protective ≫ growth, despite less headroom) and the null declarative placebo are content signals; the one-signed cash shift is not. "Content-conditional caution" vs "generic imperative-induced caution with content-modulated gain" cannot be separated without the swap arm.

3.3 Attack 2 — cash mediation: substantially correct, with two survivors

Quantity	Result
corr(Δcash, ΔMAS), ISFJ	−1.000 exactly (MAS_ISFJ ≡ 1 − mean cash; the persona never breaches its target from above)
corr(Δcash, ΔMAS), ENTJ	+0.905
Crash MDD arm effect controlling mean cash	+3.29 → −2.88 pp (sign flips; the crash benefit is entirely cash-mediated — which IS the mandate, but kills "prevents panic-selling" as mechanism)
ENTJ MAS arm effect controlling mean cash	+0.074, p≈5e-12 — SURVIVES (churn/side-of-target mediated, not cash-mediated)
Bull-trap RG arm effect controlling mean cash	−8.33, p=0.0001 — survives the mean-cash control (B), but the memory arm's RG level (78.4) sits within 1.4 points of the all-cash-HOLD baseline (79.8) (A) — partial mediation, nuance preserved (D6)
Metric inter-correlations	"Three metrics, one variable" holds fully for cash–MDD (r 0.67–0.89) and for ISFJ MAS (identity); |r(MAS,RG)| = 0.04–0.39 — MAS and RG retain partial independence. Report the matrix.

3.4 Attack 3 — prior art: verified, nobody owns it

Paper	What it owns	Why it does not own this claim
Li/Kim (COLM 2024)	System-prompt repetition swept over injection probability p, with a mandatory capability-cost axis — fully pre-empts App. H's frequency ablation	Never varies the semantic content of the injected instruction; genuine multi-turn setting
ContextEcho (2605.24279)	23-model long-horizon persona drift; length-matched filler control; an anchor-CONTENT ablation exists (identity-sentence vs format-demo)	No behavioural task, no ground truth, no content×regime sign structure; register drift, judge-scored
Meta Proactive Memory (2607.08716)	"Behavioral state decay" (the term); when-to-inject policy ablations; "unnecessary interventions can be harmful" (owns the broad C10)	Varies policy/domain, never the standing mandate's semantics in a fixed environment
CLQT (2606.29771 §3.5)	Threshold-triggered drift-warning injection inside a mandate-aware financial benchmark	Never ablates mandate content; names it as future work (§8.8)

The seam Cycle 1 identified is real but NARROWER than Cycle 1 stated: the "sign-reversal" half of the claim is metric-manufactured (Attack 2); what is unclaimed is content-modulated behavioural steering by an injected directive in a fixed, ground-truth-scored environment.

Full-corpus result (Section 8, all 47 papers read): the four-paper check above understates how much of the design logic is prior art. ContextEcho already combines a per-cell length-matched filler control, an imperative anchor at the recency position, an anchor-CONTENT ablation (App. G) and a 23-model panel — and finds the anchor lifts no-drift targets to ceiling ("compliance amplifier", §3.2 Q4), which is the most economical alternative explanation of FinPersona's memory-vs-static gap in a stateless setting where there was no decay to reverse. Ross & Lo show stateless allocations are a near-function of the stated risk label, so a louder risk label moving cash toward its pole — and an opposed-target split — is the expected outcome. Li et al. (COLM 2024) treat per-turn system-prompt repetition as a baseline. The reframed claim therefore survives as novel in DOMAIN and READOUT (behavioural allocation, opposed targets, generating-function ground truth), not in injection logic, and must be positioned that way.

3.5 The strongest defensible form of the claim

Converged formulation (B and C independently produced near-identical texts; A concurs):

"In a fixed synthetic market with stateless single-turn agents, appending an imperative behavioural directive at the recency position shifts trading behaviour toward cash/inaction in a content-modulated way — protective ≫ growth ≫ neutral in magnitude (model-level $p < 0.002$ for both endpoint personas across 18 models, $d_z \approx \pm 1.1$), while a length- and position-matched declarative placebo produces no such shift (one model), and a numerical-target-only mandate eliminates the harm in all three models tested. Scored against opposed fixed allocation targets, this one-signed behavioural shift yields the observed 17/18 vs 16/18 adherence split. Whether the content-dependence is semantic or reducible to directive force is settled by the swapped-mandate arm [new experiment]."

As "the sign of the behavioural effect reverses with mandate semantics", the contribution does NOT survive. As the statement above, it survives all three attacks and is unclaimed in the 47-paper corpus — with the qualification from the full read (Section 8) that its injection-and-control logic is anticipated by ContextEcho and Li et al., so the revision must claim novelty of domain and readout, cite ContextEcho's compliance-amplifier finding as the null it rejects (or fails to reject — P2's wrapper-only and directive-placebo arms are the test), and cite Ross & Lo as the reason the opposed-target split is expected rather than surprising.

4. Adjudicated programme

4.1 Verdicts on Cycle 1's cut list

Item	Cycle 1	Cycle 2 (unanimous unless noted)	Cost / P(null)
M1 — attention mass on mandate span across the rollout	Cut	CUT — not merely unlikely, undefined: there is no rollout; the trajectory is flat by construction. Where a multi-turn version can run, it is already published (When Attention Closes; Li/Kim).	— / P(null)≈1 by design
M4 — logit attribution for the mini-model crash reversal	Cut	CUT — triple-gated on a crash effect that fails at model level, a mini-class pattern broken by GPT-5-mini, and an LCR "mechanism" with zero cross-model validity (r=0.13, p=0.60).	— / very high
M3 — causal ablate-and-restore (Cycle 1's internal split)	Split: A keep-last; B,C cut	RESOLVED UNANIMOUSLY: cut from the critical path; permit at most as a post-P2, pre-registered appendix on one open model (random-donor + self-patch controls, null committed to publication). Decisive argument (C): after the mandatory reframe the paper no longer makes a salience claim, so the objection M3 answers will not exist; the live objection (content vs directive force) is answered by the swapped-	2–3 wk if run / high

		mandate + directive-placebo arms at ~1% of the cost, on 18 models instead of one. B adds: even a positive M3 on one 7–9B model would be dismissed as open-weight-only. A concurs on budget grounds (P(informative non-null) ≤ 0.25).	
ABIDES port	Cut, keep retraction	CUT; delete §3.1's "realism reintroduces subjectivity" (false and checkable — ABIDES/Hashimoto keep exogenous fundamentals). Porting fixes no validity problem and inherits the P/E leak.	2–3 months if run / —
Full-grid OCEAN as "validation"	Cut framing	CUT AS VALIDATION; keep the existing OCEAN data as the vocabulary-manipulation gradient it already is (O1/O2/O3).	—

4.2 The Cycle 2 ordered programme

Merged from three independently-produced programmes that agree on order and content to an unusual degree. Effort assumes one researcher plus API budget; every item states what the paper claims on success AND on failure.

#	Item	Effort / compute	P(null / adverse)	Claim if it succeeds	Claim if it fails
P0	Truth reconciliation (do FIRST, before any compute): fix the Figure-3 binning bug and regenerate on the full horizon; withdraw the 4.4× multiplier and monotone-compounding language; fix Fig 4 crash column (the corrected 14/18 IMPROVES the paper), Fig 5 signs, sign-convention harmonisation, N=263, dagger, placebo range, self-reference, App. F/H cohort provenance, temperature, SMA and Table 2 observation-space corrections; rewrite abstract/§1/§3.2 to the stateless mandate-absent reality; retitle away from MSD/psychometric stability; add the ~15-citation set with differentiation paragraphs (drafts exist in RELATED_WORK_UPDATE_AUG2026); report effect sizes, mixed-effects battery, BH correction, metric-correlation matrix, cash-mediation table, zero-trade/degeneracy audit (4 parse errors in 531,600 steps is a genuinely publishable reliability number), trivial-policy baselines (99.7/79.8/66.7).	1–2 wk, $0	~0 (arithmetic)	The paper stops being refutable in twenty minutes from its own repo; corrected Figure 3 is STRONGER (monotone ratio ~9× on the plotted statistic)	n/a
P1	Metric repair + communicated-target axis: dual-track MAS (target stated in prompt — as O3 already does — vs unstated "revealed-preference" track); RG → mandate-conditional regret with trivial-policy floor; CI renamed MDD or dropped; C_ideal reported stipulated AND JFE-derived side-by-side; EPS=V/15 sensitivity {10,15,25}; classifier retrained (static-only, held-out models, documented provenance) or all classifier claims dropped; t=0 separability baseline published as a finding (cite PERSIST/Ross & Lo).	2–4 wk, low	Low; if the P/E-rule baseline saturates RG, RG is retired honestly	A defensible metric suite plus a novel stated-vs-unstated-target contrast no neighbour has	Metrics honestly retired; the sign-count result stands on MAS alone
P2	The causal package (pre-registered, predictions stated in the paper): (i) swapped-mandate arm — ISFJ text into ENTJ agent and vice versa; B argues full-grid 18 models × flat × 5 seeds ≈ 180 sims since	3–4 wk, ~500–2,000 cheap sims	Informative either way; ~30% the directive placebo	"Behavioural shift tracks injected content, not persona	"Any imperative at the recency position shifts stateless agents toward inaction; content

	it is THE result either way, C accepts ≥3 models — run full grid, it is cheap; (ii) directive placebo (imperative-, delimiter-, length-matched, behaviourally irrelevant); (iii) wrapper-only arm ("ACTIVE MEMORY REFRESH" framing with no mandate — the wrapper is itself an untested imperative, V9); (iv) wording dose-response (paraphrase, abbreviation, imperative strength).		absorbs most of the effect — an uncomfortable but publishable finding	identity or directive force" — the COLM-main claim, unclaimed in the corpus	modulates magnitude only" — findings/workshop, and better learned now than in review
P3	Coverage for power, NOT seeds: add the 30–70B open-weight tier with full behavioural metrics (N_models ≈ 24–26); foreground the Qwen language/behaviour dissociation; re-test the crash contrast at model level on the enlarged suite. (C's argument, endorsed: effective N is models — 20 seeds of the same 18 models cannot rescue a model-level p≈0.10–0.24.)	3–4 wk, moderate	Moderate	Either the crash effect reaches honest model-level significance, or it is buried with a power analysis; ISFJ-scoped claim gains precision	Crash axis remains a benchmark scenario, not a finding
P4	(Optional) Method deliverable: behavioural-trigger adaptive injection (|C_t − target| threshold), benchmarked against CLQT's Ct<0.7 and Li/Kim's SPR-p, with a mandatory capability-cost axis.	2–3 wk, moderate	Moderate ("no better than baselines")	The one "method" contribution	Reported as a negative comparison
P5	(Optional appendix) One pre-registered M3-style patch on one open model, only after P2, null committed to publication.	2–3 wk	High	Mechanistic garnish	A published null, as committed

4.3 Addendum — authors' decision on the mechanistic programme (22 Aug 2026)

After reading this report the authors stated that (a) the mechanistic-interpretability experiments are retained, and (b) the synthetic environment and agent are to be reworked, so the current implementation is not a constraint on the revision. The reviewers' "cut M1/M3/M4" verdicts above were reached against the CURRENT architecture (single-turn, mandate-absent static arm), under which those experiments measure a decay that cannot occur. They are not verdicts against mechanistic work as such. Two consistent ways forward are laid out in docs/FinPersona-Bench_Position_and_Revision_Plan_Update_Aug2026: Path B — rebuild the agent with genuinely accumulating context and the mandate present in every arm, decouple elapsed context from market phase in the reworked environment, keep the current grid as the stateless control arm, and run M1–M4 as the plan designed them, pre-registered to report the accessibility-vs-behaviour dissociation that Li 2024, When Attention Closes, Menon 2026, Dongre 2025 and Kruthof 2026 jointly predict; Path A — keep the stateless architecture and re-aim M1–M4 at the position, content and mediation of the injected directive. Under Path B, GO condition 2 below ("no mechanistic-interpretability spend this cycle") is superseded by: mechanistic spend only after the stateful agent and the phase-decoupled environment exist, and only under pre-registration. Conditions 1, 3–10 stand unchanged on both paths. The literature findings (Section 8) and the verification findings (Sections 1–2) are unaffected by the choice of path.

5. Per-reviewer findings (preserved separately)

Full reports are preserved verbatim in docs/cycle2_reviews/. Below: each reviewer's distinctive contributions and errors — including where a reviewer was wrong, because that calibrates how much to trust convergence.

5.1 Reviewer A — forensic auditor (reviewer_A_forensic.md)

- Re-derived both registers in full from 531,600 decision rows; verified Tables 1/9/10/11 reproduce to 3–4 decimals; found the binning bug independently and showed the corrected cumulative statistic gives ~9× while the quartile-local gap collapses at Q4.
- Distinctive new findings: the crash δ-sweep varies the price-path MDD floor by only 0.6 pp (App. J near-vacuous); LCR has no cross-model validity (precision 0.50, recall 0.33, p=0.60); mandate re-injection raises cash for all three personas; RG's bull-trap gap ≈ the metric's mechanical cash penalty (memory arm within 1.4 points of the all-cash baseline); personas not separable at t=0; App. H's Claude raw data absent from the repo; parse failures 4/531,600; 306/2,658 runs traded zero times.
- Errors found in A's report (orchestrator adjudication): (i) claimed Figure 4's ISFJ crash 2/18 "reproduces exactly" — it reproduces only under the same sign inversion that produced the figure error; true value 14/18. (ii) Consequently its F-NEW-8 (rebuttal 14/14 claim "contradicted") overstated: the rebuttal is roughly right (12/14 API models) and the figure is wrong. (iii) Its Error-#7 τ1 undervaluation figure (~20%) is the all-days number; the τ1 number is ~50%. (iv) Repeated Cycle 1's "mandate once vs twice" description of the arms, which is wrong (mandate absent vs present).
- Verdict: NO-GO for anything resembling the current draft; conditional GO on the reframed prompt-composition/refraction paper.

5.2 Reviewer B — author's advocate (reviewer_B_advocate.md)

- Strongest single point: the paper's NUMBERS are overwhelmingly faithful — Tables 1, 9, 10, 11 and Apps E, F, G, L reproduce digit-for-digit from raw CSVs. The forensic risk lives in figures and framing, not in the data. Three of the figure errors (Fig 4 crash, Fig 5 crash, the Fig-3 truncation making the trajectory look weaker than the corrected monotone ~9×) actually run AGAINST the authors' interest.
- Distinctive contributions: found Figure 4's crash column is wrong and computed the true 14/18 (confirmed by orchestrator); the ISFJ-scoped crash rescue (p=0.0038); the model-level interaction p=3.8e-5 and dz ≈ ±1.1; the defence of Error #4/#5 ("arithmetically impossible" is false — mislabelled axis); the refutation of Error #7 (τ1 undervaluation 46–56%); the App-J severity-trend n.s. defence (p≈0.16); the reconciliation of the Sonnet cohorts including exact T=800 verification; the observation that the corrected Figure 3 is monotone with ratio 9.0 under the plotted statistic (Cycle 1's "peaks at Q3" is wrong for that statistic).
- Errors/limits in B's report: repeated the "mandate once (system) vs twice" characterisation of the arms — wrong; the static system prompt contains no core mandate (it contains only the descriptive persona extension). B's O3-as-anti-tautology reading is contested by C (divergence D5, preserved).
- Verdict: GO, conditional on the reframe, the swap/directive-placebo arms, ISFJ-rescoping of the crash claim, and the cuts.

5.3 Reviewer C — area chair (reviewer_C_areachair.md)

- Distinctive contributions: discovered the static arm never receives the core mandate (V1b; orchestrator-verified — the single largest new architectural fact of the cycle); the observation that Eq. 3/5 are consistent with the code while the paper's prose is not; the degenerate-success audit (35.6%

of ISFJ-memory flat runs and 47% of crash runs execute zero trades — "perfect adherence" = inert agent); the no-cash-target regex finding (MAS grades distance to a number the agent is never told); Table 7's flat-only caption error; the accept/reject package definitions; the argument that seeds cannot buy model-level power but models can.

- C's O3 reading (completes the tautology) opposes B's (anti-tautology card) — preserved as D5. C's bucket-END computation (peaks Q3 at 6.25×) initially looked like a conflict with A/B's monotone bucket-means; orchestrator reproduced both — different statistics, no substantive disagreement (D3).
- Verdict: GO — conditionally, with six binding conditions (adopted in Section 9); explicit reject-package tripwires.

5.4 Literature readers L1–L3 (litreview_part1/2/3.md)

- L1 (related_work_2026, first 14; 406 pp): identified Arike et al. 2025 as a direct, uncited competitor (behavioural goal drift of LLM traders, 4 models × 20 seeds, >100K tokens; imperative elicitation reduces drift; token-distance ablation shows distance is NOT the driver); Dongre 2025 (bounded equilibria) and Laban 2025 (degradation from turn 2, variance-dominated) as contradicting "compounds"; Mehri 2026 App. E as prior per-turn verbatim goal re-injection with per-model reversals; Song 2026's lexical-transparency mechanism as the better explanation of the O3 result. Disagreed with Cycle 1 on C10 (Mehri/Dongre anticipate it) and C11 (Arike's ablation is not a placebo). ERROR by L1 (orchestrator-refuted): flagged Table 5 as possibly printing cash fractions — the values are MAS deviations (MAS_ISFJ = 1 − cash; placebo cash 5.0% ↔ 0.950 exactly), so memory's 0.730 IS the improvement the text describes.
- L2 (related_work_2026, last 14; 549 pp): verified CLQT §3.5/§8.8, When Attention Closes App. D.5/H, Meta's Table 2, and Jiang/Peng/Yan's coefficients verbatim; found Menon's frontier zero-drift-over-30-steps result and instruction-hierarchy test (user-turn directives override system goals for some families — the mechanism behind a recency directive and its model-dependence); Kocielnik App. I.9 (any in-context text shifts risk-taking, content-modulated — a caution on reading the placebo as a clean null); CLQT's reliability-driven HOLDs (report parse/refusal/always-HOLD before attributing cash shifts to content); the Always-On survey's episodic/always-on vocabulary. Corrected four Cycle 1 wording-strength claims.
- L3 (mech_interp + gap_sweep, 19 items; ~370 pp + 2 HTML): found the "Kim_Suzgun" file is Li et al. (COLM 2024) and that Li's attention decay is strictly cross-turn with a within-turn plateau (predicts NO decay in a stateless prompt); verified ContextEcho's six facts and surfaced its compliance-amplifier and user-turn-beats-system-prompt findings; verified Ross & Lo's 57–88% and Kruthof's 8–99%; refuted Cycle 1's "opposite capacity ordering"; found ROME mis-cited in the revision plan (causal tracing, no DLA) and Hong mis-cited in the paper's intro; mapped M1–M4 feasibility against the mech-interp corpus (M1 flat by mechanism; M3 only as a mediation test; Arghal's 0/456 single-layer patching). Took the hardest line: the reframed claim is "anticipated in form".

6. Convergence and divergence analysis

6.1 Convergence (three independent reviewers, no shared context)

Twelve points reached independently by all three — with the caveat that all three are AI reviewers with correlated priors from the same materials, convergence on recomputed NUMBERS (which agree to the decimal) is strong evidence; convergence on judgements is weaker evidence:

1. The Figure-3 binning bug (bins=[0,25,50,75,100]) as the root cause of the 4.4× and of Errors #4/#5/#21 — each reviewer reproduced every figure annotation to the digit.
2. No conversational context accumulates; portfolio state is passed every step; Appendix L's dollar quotes are the prompt echoed back.
3. The pooled crash headline fails every defensible model-level specification (p = 0.096–0.238).
4. The 17/18 and 16/18 counts verify exactly and pass model-level tests — the only headline numbers in the paper that do.
5. Re-injection raises cash for ALL three personas (+25.9/+9.6/+5.9 pp); the metric sign reversal is manufactured by target placement; content survives in the magnitude ordering.
6. The classifier is trained on both arms, days 1–5, with a within-run validation split — circular; all P(intended-persona) numbers quarantined.
7. Personas are not behaviourally separable at t=0; no persona ever approaches its C_ideal; static ENTJ drifts toward its target.
8. The Sonnet 0.903/0.962/0.988 values are different cohorts, not one irreconcilable number; the paper's failure is nondisclosure; App. H's leg is unverifiable from the released artifact.
9. Crash N = 263, not 270 (Gemma-3-4B incomplete — the undefined dagger).
10. Prior art: none of Li/Kim, ContextEcho, Meta, CLQT owns the content-modulated steering claim; Li/Kim fully pre-empts the frequency ablation.
11. Programme: M1/M4/ABIDES cut; M3 off the critical path; the swapped-mandate + directive-placebo arms are the decisive, cheap discriminator; no new compute before the rewrite.
12. GO/NO-GO: conditional GO on the reframed paper; NO-GO if the MSD-decay/4.4×/re-grounding framing or pooled-N inference is retained.

6.2 Divergences, and what the adjudications found

#	Divergence	Positions	Orchestrator adjudication
D1	Does the static arm carry the mandate?	C: mandate absent from static arm. A & B: repeated Cycle 1's "once vs twice" framing.	C is right. Verified directly: get_financial_persona never touches core_mandate; the static system prompt = persona baseline + descriptive financial_extension only. Both Cycle 1 AND two of three Cycle 2 reviewers propagated the wrong description — a caution about echo effects in AI review. Nuance: the static arm is not persona-free; the extension carries semantically aligned dispositions, so "mandate-absent", not "unconditioned".
D2	Figure 4's ISFJ crash cell	A: 2/18 "reproduces exactly". B: printed 2/18 is a production error; true 14/18.	B is right. Orchestrator recomputation: improves 14, worsens 2, exact ties 2. The printed crash column equals the WORSENS counts (ISFJ 2, INTJ 10 both match; ENTJ 9 is a 9–9 tie) — the column is sign-inverted, same family as Figure 5. A matched the printed number by making the same sign error the figure did.
D3	Shape of the corrected 4.4× trajectory	A/B: monotone, ratio ~9×. C: peaks at Q3 (6.25×), falls to 4.57×.	Both right — different statistics, both reproduced by the orchestrator: bucket-MEANS of the expanding-min gap (what Figure 3 plots): 0.0044/0.0112/0.0340/0.0391, monotone, ratio 9.0. Bucket-ENDS: 0.0072/0.0181/0.0450/0.0329, peaks Q3. Quartile-LOCAL

			gaps collapse at Q4 (A: 0.0048; B: 0.0002 under a stricter reset — definitional variants agreeing qualitatively). Consolidated statement: on the paper's plotted statistic the corrected figure is monotone ~9×; the divergence is phase-locked (grows through scripted panic, stops/reverts in scripted stabilisation), so "compounds over time" remains unsupported. NOTE: Cycle 1's own quartile numbers (0.0100/0.0245/0.0577/0.0424) were reproduced by NOBODY and are superseded.
D4	Error #7 (bull-trap undervaluation)	A: confirmed with correction (P<V ~20% of days). B: refuted (46–56% of τ1 days).	B is right for the claim at issue. Orchestrator: P<V on 50.0% of τ1 days (46–56% by seed), 0.7% in mania, 0.0% in blow-off, 20.2% over all days. Cycle 1's "App. A guarantees the asset is never undervalued there" is false; A's 20% was the all-days number. §4.3.1's sentence is possible as written; the deeper RG problem (memory RG ≈ all-cash baseline) stands regardless.
D5	What the O3 ablation means	B: the best anti-tautology card (stating the numerical target removes the harm ⇒ trait language drives it). C: completes the tautology (the injected text states the graded answer; of course adherence is perfect).	GENUINELY UNRESOLVED — an interpretive, not numerical, disagreement (the O3 numbers themselves verify: −0.028/−0.038/−0.093, all three models). Both readings predict different swapped-mandate outcomes, which is exactly why P2 is the decisive experiment. Preserved, not flattened.
D6	Is RG cash-mediated?	B: arm effect survives controlling mean cash (p=0.0001). A: the RG gap ≈ the metric's mechanical cash penalty on τ1 dip days; memory arm within 1.4 pts of the all-cash baseline.	Both computations verified; they answer different questions (mean-cash regression vs level-vs-baseline). Consolidated: RG is PARTIALLY cash-mediated — it survives a mean-cash control but its level is nearly saturated by a trivial cash policy. Report both.
D7	App. J's mildest-crash inversion	A: real, mechanism = denominator artifact on a near-vacuous δ-sweep. B: not significant (p≈0.16), Cycle 1 interpreted noise. C: confirmed numerically, unremarked in paper.	All numbers agree (−15.0/−16.5/−18.8%). Consolidated: the trend is statistically noise (B) AND the sweep varies the environment by only 0.6 pp (A) — App. J is weak evidence for anything and should be demoted either way.
D8	Crash rescue scope	B: ISFJ-scoped claim survives (p=0.0038). A: did not test per-persona scoping (and its F-NEW-8 pointed the wrong way via the inverted figure). C: pooled-only analysis.	B's rescue confirmed by orchestrator (14/18, +7.47 pp, Wilcoxon p=0.0038; sign 14/16 excl. ties, p=0.0042). Mandatory caveat from C's audit: the mechanism is market-avoidance (zero-trade runs), not panic-selling prevention.
D9	Sonnet 0.903 provenance detail	A/B: March pilot cohort. C: harness-vs-per-step pipeline difference.	Consistent, not conflicting: 0.903 is the per-step-pipeline recompute OF the March pilot cohort (april_baseline.json names the source directory). Resolved.
D10	Is the reframed claim novel? (literature readers)	L1, L2: unclaimed in their shares, with partial anticipations (Arike, Mehri, Dongre, Hashimoto; Kocielnik, Menon, Wu). L3: "anticipated in form" — ContextEcho + Arditi + Li own the directive-at-recency + length-matched-control + content-ablation logic; Ross & Lo make the outcome expected; residual novelty is the allocation readout in a synthetic market.	Same facts, different novelty thresholds; L3's is the one a top-venue reviewer with ContextEcho open will apply. Consolidated: novel in DOMAIN and READOUT, not in injection logic. The revision must say so, and must treat ContextEcho's compliance-amplifier finding as the null hypothesis. Not flattened: L1/L2's "unclaimed" is defensible for the 17/18-vs-16/18 result as a specific empirical fact; L3's is the correct framing for the contribution claim.

Meaning of the divergences: every NUMERICAL disagreement dissolved under adjudication into either a definitional difference (D3, D6, D9), a denominator difference (D4), or an outright error by one reviewer (D1, D2) — and in both outright-error cases the error was an echo of an earlier report's framing (Cycle 1's "once vs twice"; the figure's own sign convention). The one genuinely open disagreement (D5) is interpretive and is exactly the question the P2 experiment is designed to answer. Practical lesson: claims that merely repeat a prior report's characterisation should be weighted near zero; only recomputed numbers carried this cycle.

7. Orchestrator verification record

Independent computations performed at consolidation (not delegated), from agent/ source and the cached run-level data (scratchpad runs_metrics.csv / allsteps.parquet, rebuilt from results_april and results_may):

- Static-arm mandate absence: read static_agent.py (loads financial_extension only, line 50), memory_agent.py (loads core_mandate, line 57; injects at lines 99–105), prompts.py get_financial_persona (no mandate reference), and all 16 personas in mbti_profiles.json. Confirms C, refutes the "once vs twice" framing.
- Figure 4: read the rendered page (grid ISFJ 17/2/7, INTJ 11/10/5, ENTJ 2/9/4). Recomputed all nine cells: flat column = MAS improve counts (17/11/2 — exact match); bull column = RG-higher counts (7/5/4 — exact match); crash column = MDD WORSENS counts (2/10, ENTJ tie) vs true improves (14/8/9). Crash column sign-inverted; root cause established.
- ISFJ crash: per-model MDD gaps listed; improves 14, worsens 2 (Gemma-3-4B −11.2, GPT-5.4-mini −1.0), ties 2 (DeepSeek, Qwen: both arms zero drawdown — all-cash both arms); mean +7.47 pp; Wilcoxon $p=0.0038$; sign 14/16, $p=0.0042$.
- Bull-trap undervaluation by phase: legitimate_rise (days 1–80) P<V 50.0% (46.2–56.2% by seed); mania 0.7%; blowoff 0.0%; all days 20.2%.
- 4.4× trajectory, both statistics, 18 models at $\delta=0.92$: bucket-means 0.0044/0.0112/0.0340/0.0391 (ratios 1/2.56/7.82/9.0); bucket-ends 0.0072/0.0181/0.0450/0.0329 (ratios 1/2.51/6.25/4.57). Matches A/B and C respectively to the fourth decimal.
- Interaction statistic: per-model flat MAS gaps, ENTJ (+0.126) vs ISFJ (−0.259), Wilcoxon $p=3.81\text{e-}5$. Matches B.

Items no reviewer could verify, reported as UNRESOLVED rather than guessed:

- App. H's Claude injection-frequency raw data (the 0.962/0.901 leg) — absent from the released repository.
- Whether the prompt's "SMA60" series is a 50- or 60-day average (B and C read the code differently; not re-adjudicated — either way prompt and paper disagree, which is the reportable fact).
- The provenance of Cycle 1's quartile-gap numbers (0.0100/0.0245/0.0577/0.0424) — reproduced by nobody; superseded by three mutually consistent recomputations.
- When Attention Closes App. D.5's patching nulls (cited by Cycle 1 in the M3 dispute) — C flagged it as not re-read; subsequently read in full by L2: the null is real (n=30 pairs, Mistral-7B, fact recall, one layer block) but the authors themselves call it "descriptive localization, not causal identification" — a risk signal for M3, not a settled null. Resolved.

Literature verification record:

- The orchestrator did not read the 47 PDFs directly; each was read in full by exactly one of three literature readers (L1/L2/L3), who wrote per-paper assessments before opening Cycle 1's table.

Targeted verifications Cycle 1 depended on were assigned explicitly and all returned: CLQT §3.5 (C_t<0.7 trigger) and §8.8 (mandate variation as future work) — verbatim; Meta's "behavioral state decay" definition — verbatim, but always-on injection never falls below baseline in its Table 2; Jiang/Peng/Yan $R^2 \approx 0.05$ and Neuroticism largest — confirmed, "5–8 pp" is derived (4.8–8.7 pp) not stated; Li et al.'s SPR p-sweep with MMLU axis — confirmed, authorship corrected; Ross & Lo 57–88% and input-sensitivity criterion — verbatim; Kruthof KBV 8–99% — verified, "opposite capacity ordering" not supported; ContextEcho 23 models / 9,643–9,716 turns / filler arm / content ablation / mode sign-flip / judge-free metrics — all six verified.

- One literature-reader claim was refuted by the orchestrator against the run data: L1's suggestion that App. F Table 5 prints cash fractions rather than MAS deviations. MAS_ISFJ ≡ 1 − mean cash (Reviewer C, exact); the placebo cohort's ISFJ cash of 5.0% maps to 0.950, matching Table 5; memory's 0.730 is therefore the improvement the text reports. Table 5 is correctly labelled.

8. Full-corpus literature review (47 papers, read in full)

Method. The 47 items in references/ were split across three independent readers — L1: related_work_2026 items 1–14 (406 pp); L2: related_work_2026 items 15–28 (549 pp); L3: mech_interp (8, incl. two HTML reports) + gap_sweep_aug2026 (11) (~370 pp + 2 HTML). Every page was extracted with PyMuPDF (HTML via html.parser) and read; each reader first read the paper under review, then wrote per-paper assessments (citation, what it does, relevant findings with section/page, which ORIGINAL FinPersona claims it anticipates/contradicts/supports, bearing on the REFRAMED claim, citation recommendation) and only then opened Cycle 1's 19-claim table to record agreement/disagreement. Full per-paper assessments: docs/cycle2_reviews/litreview_part{1,2,3}.md.

8.1 Corpus map — one row per paper

Relevance: HIGH = bears directly on a FinPersona claim or its interpretation; MED = positioning/partial; LOW = tooling or tangential; NONE = irrelevant. "Cite" = MUST / yes / optional / no.

#	Paper (venue)	Rel.	What it establishes that matters here	Bearing on REFRAMED claim	Cite
1	Byrd et al. 2019, ABIDES (arXiv)	LOW	Latent fundamental + noisy observables as a simulation principle; no LLMs. Not an existence proof for LLM-agent synthetic markets (Hashimoto is).	none	optional
2	Abdelnabi et al. 2025, task-drift activation deltas (SaTML)	MED	Instruction-bearing text appended at the END shifts internal task representation most; instruction-free text does not (Table VI). Mechanistic precedent for imperative-vs-declarative.	supportive	yes
3	Arike et al. 2025, goal drift in LM agents (MATS/Apollo)	HIGH	DIRECT COMPETITOR, uncited: LLM trading sim, behavioural drift scores incl. drift-through-inaction, 4 models × 20 seeds, up to 74 steps / >100K tokens; imperative elicitation reduces drift model-dependently; token-distance ablation: distance from system prompt is NOT the driver (§5.1).	partially anticipates (imperative directive moves trading behaviour content-consistently)	MUST
4	Laban et al. 2025, LLMs get lost in multi-turn	MED	SNOWBALL per-turn repetition partially repairs degradation; degradation present from turn 2, variance-	none direct	yes

			dominated — not compounding. Loss-in-middle-turns = recency premise.		
5	Mehri et al. 2026, goal alignment in user simulators (TACL)	MED-HIGH	Per-turn goal re-injection (App. E) improves alignment with model- and dimension-specific REVERSALS (Qwen-7B profile 88.3→70.1) — prior "re-grounding not universally beneficial".	partial	yes
6	Tosato et al. 2025, PERSIST (AAAI 2026)	MED	Persona self-report instability; SD>0.3 at T=0 from question order alone → FinPersona's n=3–5-seed ablations sit inside this noise band.	tangential	yes
7	Song et al. 2026, questionnaires mischaracterize LLMs	MED-HIGH	Persona prompts move questionnaire answers (lexical transparency) but not realistic generation. FinPersona's O3 result (vocabulary removed → ENTJ penalty vanishes) fits the cue account; undercuts "framework-independent".	predicts content-modulation sign	yes
8	Chen et al. 2026, AlphaSAGE (ICLR 2026)	NONE	Formulaic alpha mining; irrelevant.	none	no
9	Dongre et al. 2025, Drift No More (AAAI-W 2026)	HIGH	Multi-turn drift reaches a bounded, noise-limited equilibrium; reminders at t=4,7 lower it model-specifically (Llama-70B D* RISES). Contradicts "compounds over time".	partial	MUST
10	Hashimoto et al. 2025, ABM financial market with LLMs	MED-HIGH	Stateless single-turn LLM buy/sell prompt with imperative allocation directives in a LOB market with hidden fundamental; Qwen outlier. The LLM-agent synthetic-market existence proof (not ABIDES).	partial (stateless directive shapes allocation)	yes
11	Buakhaw et al. 2025, Deflanderization (CPDC)	LOW-MED	Defines flanderization; persona emphasis degrades task execution — consonant with ENTJ over-trading.	weak	optional
12	Fan et al. 2025, AI-Trader (live benchmark)	MED	Strong model-dependence incl. passivity/cash-holding as a spontaneous default ("hold-to-die") — a confound for reading ISFJ 17/18 as mandate-driven; report the static cash baseline.	caveat	yes
13	Saini et al. 2026, gradient-ascent persona control (TMLR)	LOW	Manual directives have modest, model-dependent effects; token-priming via induction heads — relevant to rationale-vocabulary echo.	tangential	optional
14	Zhang et al. 2026, Locate-Steer-Improve survey	LOW	Catalogue of persona-vector work; cite primaries instead.	none	optional
15	Menon et al. 2026, inherited goal drift (ICLR-W)	HIGH	Frontier models show ZERO drift over 30 steps under adversarial pressure in Arike's trading env (drift needs conditioning on a weaker model's trajectory); instruction-hierarchy test: user-turn instructions override system goals for Gemini-2.5-Flash/Claude-Sonnet-4.5 but not GPT-5 family — the mechanism for a recency directive and its model-dependence; ER triage executed.	mechanistic support + model-dependence	MUST
16	Cheng et al. 2026, representation steering of refusal	NONE	Refusal-circuit tooling only.	none	optional
17	Hu & Zhao 2026, Fin-Bias	LOW-MED	Single-shot: an in-context analyst opinion shifts financial decisions	weak	yes

			toward its content (herding 5–10 pp; fake ratings ~30%), size-independent.		
18	Dongre et al. 2026, When Attention Closes	HIGH	Attention to system-prompt goal tokens declines 27–48% over 50 turns while recall stays 100% (accessibility decay WITHOUT behavioural decay); App. H: prepended periodic re-injection with a length/entity-matched irrelevant block — null for both arms, qualitative; App. D.5 patching null (n=30, one model, one task; "descriptive, not causal").	caution; prior analogous 3-arm design	MUST
19	Moskvoretskii et al. 2026, persona vectors in pretraining	NONE	Tooling/background only.	none	optional
20	Xia et al. 2026, Agentic Trading survey	LOW	Positioning; MR-4 (execution costs) exposes FinPersona's frictionless W_t.	none	yes
21	Zhu et al. 2026, KTD-Fin	LOW-MED	Memorisation is real and in-prompt instructions cannot suppress it; agents default to cash/no-trade when handles are masked (baseline disposition toward inaction).	background	yes
22	Kocielnik et al. 2026, rethinking psychometric eval	HIGH	Big Five reliable but non-predictive of within-model behaviour (undercuts OCEAN-as-validation); persona prompting moves self-reports not behaviour; App. I.9: ANY in-context self-report text shifts risk-taking, and policy framing shifts it content-modulatedly (loss-averse 8.1 vs gain-seeking 18.9 cards).	closest behavioural analogue; caution on placebo-as-null	MUST
23	Guda 2026, agent security / regulatory	NONE	Motivational only.	none	optional
24	Qu & Chen 2026, CLQT	HIGH	Closest BENCHMARK competitor. §3.5 $C_t<0.7$ drift-warning trigger — verified verbatim; a_{target} undefined; §8.8 mandate variation = future work — verified; coherence gap +0.30/+0.23; style drift ≈0 over 26 rounds; 44–50% HOLD rates were reliability artefacts (schema failure, tool-turn exhaustion) — FinPersona must report parse/refusal/always-HOLD rates before attributing cash shifts to content.	does not own; two cautions	MUST
25	Ding et al. 2026, Always-On Agents survey	MED	§2.1: an agent that resets state between requests is EPISODIC — FinPersona's agents are episodic, not always-on; §6.4.3 baseline-beats-memory (multiply replicated); TriggerBench (prospective recall degrades); "recall is not compliance".	background	yes
26	Ko & Geiping 2026, attractor states	LOW-MED	Assigned stances decay toward model-specific attractors in 20-turn debates — alternative explanation for per-model heterogeneity.	none	yes
27	Wu et al. 2026, Meta Proactive Memory	HIGH	Defines "behavioral state decay" (anticipates C1 in substance, not verbatim); selective > always-on on macro average only; always-on NEVER falls below the no-memory baseline in	weak	MUST

			Table 2 (corrects Cycle 1's gloss); ungrounded injection and an untrained memory agent are what hurt.		
28	Jiang, Peng & Yan 2024, JFE	MED	Adj. $R^2 \approx 0.05$ confirmed; Neuroticism largest for total/retirement equity (−1.74 pp/point); "5–8 pp swing" is DERIVED (4.8–8.7 pp), not stated; headline is "Neuroticism and Openness". Supports direction of OCEAN mapping, contradicts the 80-pp C_ideal spread; MBTI omits Neuroticism.	none	yes, carefully
29	Li et al. 2024, instruction (in)stability (COLM 2024) — file misnamed "Kim_Suzgun"	HIGH	System-Prompt Repetition swept over injection probability p with an MMLU cost axis — fully pre-empts App. H; attention decay is CROSS-TURN with a within-turn plateau (§4.2) → predicts NO decay inside a stateless prompt; split-softmax as an attention lever. Not in the paper's reference list.	partial (method)	MUST
30	Arditi et al. 2024, refusal direction (NeurIPS)	MED	Diff-in-means + all-layer directional ablation; adversarial-suffix analysis with a RANDOM length-matched suffix control — mechanistic precedent for placebo logic; system-prompt sensitivity is family-dependent.	methodological precedent	yes (M2/M3)
31	Chen et al. 2025, Persona Vectors	MED	Fig. 4: system-prompt content is strongly encoded at the last prompt token before generation (r=0.75–0.83) — the mandate is not "lost" in a single call; within-condition r 0.25–0.81 → per-step projection signal noisy.	consistent	yes
32	Hong et al. 2025, Context Rot (Chroma)	LOW	Retrieval/copy tasks at 10k–100k tokens; MIS-CITED in the paper's intro for instruction following; explicitly offers no mechanism.	none	reword
33	Liu et al. 2024, Lost in the Middle (TACL)	LOW-MED	U-shape: PRIMACY and recency both privileged for ≥13B — does not predict a system-prompt mandate is under-weighted; fact retrieval, not instruction following.	weak	keep, reword
34	Meng et al. 2022, ROME (NeurIPS)	LOW	Causal tracing only — NO direct logit attribution (revision plan mis-cites it for DLA).	none	only if M4
35	Templeton et al. 2024, Scaling Monosemanticity	LOW	SAEs on proprietary Claude 3 Sonnet; gradient×feature attribution is the DLA-style tool; few-shot-vector vs SAE caveat → a single "adherence direction" may be blunt.	none	only if SAEs run
36	Xiao et al. 2024, Attention Sinks (ICLR)	LOW-MED	Attention to initial tokens is large and does NOT decay with distance; any M1 must exclude the first ~4 sink tokens and use a depth-matched control span (the end-of-prompt placebo span is not one).	none	yes (M1)
37	He et al. 2024, Multi-IF (Meta)	MED	Per-turn instruction degradation is FRONT-LOADED (IFR larger 1→2 than 2→3; o1-preview 0.877→0.707) — not compounding; accumulating context AND constraints.	none	yes
38	Everitt et al. 2025, goal-	MED	Goal-directedness ≠ capability	weakly consistent	yes

	directedness (DeepMind)		(FinPersona never measures capability); App. E: degradation NOT explained by context length; motivational system prompts shift behaviour only partially.		
39	Bao et al. 2026, DRIFT-BENCH (Notre Dame)	NONE	Task drift under faulty user inputs; name collision with Kruthof.	none	no (footnote)
40	Arghal et al. 2026, behavioural+representational goal-directedness (ICML)	MED	Framework precedent for M1–M4; linear probes weak (39% vs 70% MLP); single-layer patching changes action 0/456 times, all-layer works.	none	yes if M-series
41	Sawant 2026, high-stakes personalization (position)	LOW	Informal: stateless assistants anchor on recent evidence; no data.	consistent	optional
42	Saxena et al. 2026, Machine Spirits	MED-HIGH	15 LLMs in Hommes-style markets with a DISCLOSED fundamental and endogenous price; model-specific bubbles; macro behaviour robust to memory 0/2/4 (Table 6).	none	yes
43	Ross & Lo 2026, One Size Fits None (MIT)	HIGH	Stateless LLM allocations follow a single heuristic — stated risk tolerance carries 57–88% of predictive weight (verified); input-sensitivity criterion (verified). Makes a content-modulated cash shift against opposed targets the EXPECTED outcome; offers a surrogate-model tool to report the directive's share of predictive weight.	strongly bears — partially anticipates	ESSENTIAL
44	Kruthof 2026, DriftBench (TUM)	HIGH	Knows-but-violates 8–99% with 97.3% recall (verified); rejects the forgetting/decay framing; checkpoints/monitoring reduce KBV only partly; NO capacity ordering (within-provider larger is better; Sonnet 4.6 the outlier). In a stateless design non-compliance is KBV by construction → rename the construct.	consistent; renaming	ESSENTIAL
45	Canaverde et al. 2026, SEQUOR	MED	50-turn constraint decay (averages −26/−38/−27%; Cycle 1 quoted the best-model abstract figures); re-statement resets compliance; decay occurs even with the constraint in the last 15 turns (not context loss).	consistent	yes
46	Cai et al. 2026, PushBench	LOW-MED	Generic checklist reminder: zero success delta; external verifier state helps — reminder ≠ fix; supports an externally-triggered controller design.	tangential	optional
47	Ding et al. 2026, ContextEcho (Accenture)	HIGHEST	Closest DESIGN analogue: per-cell length-matched filler control; imperative A-anchor as a USER turn at the recency position restores register across 23 models INCLUDING no-drift targets ("compliance amplifier" — the null hypothesis for FinPersona's memory arm); anchor-CONTENT ablation (App. G); user-turn beats system-prompt placement (App. I); mode-dependent sign flip; judge-free S2 compliance (so Cycle 1's "no behavioural task" is partly wrong).	ANTICIPATED IN FORM; novelty only in domain/readout	ESSENTIAL

8.2 What the corpus anticipates (consolidated across the three shares)

- The MSD CONCEPT (instructions present in context yet no longer governing behaviour): Wu et al. ("behavioral state decay"), When Attention Closes (GAR), TriggerBench via the Always-On survey, Dongre 2025 (explicit dynamical formalisation with intervention term), Li et al. (instruction drift). The name is new; the idea is not — and every prior instantiation has genuinely accumulating context, which FinPersona's arms do not (they are "episodic" in the Always-On survey's vocabulary).
- Behaviour-vs-language dissociation: CLQT's coherence gap (+0.30/+0.23, held-out judge, live-replicated), Kocielnik (persona prompts move self-report not behaviour), Song (lexical transparency), Kruthof (KBV with 97% recall), "recall is not compliance" (Always-On survey §8.2.3).
- "Re-grounding / memory is not universally beneficial" in the broad form: Always-On survey §6.4.3, CLQT, Wu et al., Mehri App. E, Dongre 2025, Laban SNOWBALL, Kruthof checkpointing, PushBench checklist null, ContextEcho mode sign-flip. Settled; what is not owned is a content-driven sign structure against opposed targets in one readout.
- Per-step re-injection as a method, and the frequency axis: Li et al. SPR p-sweep with MMLU cost (App. H fully pre-empted); Mehri every-turn reminder; Dongre sparse reminders; Wu always-on vs selective.
- Length-matched control for injected text: ContextEcho (filler arm in every cell, 41,921 evaluations), When Attention Closes App. H (length/entity-matched block, prepended, qualitative, null), Arditi (random suffix). The LOGIC is prior art; FinPersona's single-model 15-pair declarative placebo is the weakest instance in the corpus.
- Content ablation of the injected text: ContextEcho App. G (identity sentence vs format demo vs combined vs two-shot).
- Selective / event-triggered re-grounding: Wu et al. built it; CLQT ships a $C_t<0.7$ trigger (verbatim). FinPersona's Limitations list this as untested.
- Model-dependence: universal — ContextEcho (Δ −0.15…+1.00 across 23), Kruthof (8–99%), Menon (family-specific instruction hierarchy), Kocielnik, Ko & Geiping, SEQUOR, Machine Spirits, Multi-IF, AI-Trader.
- Behavioural measurement of goal adherence in an LLM TRADING simulation over a long horizon, with drift-through-inaction and imperative elicitation: Arike et al. 2025 (uncited direct competitor); Menon re-runs it with frontier models.
- Objective ground-truth market for many LLMs: Machine Spirits (disclosed fundamental, endogenous price); LLM-agent synthetic market with hidden fundamental: Hashimoto. Residual novelty: hidden V as a SCORING criterion for single agents under mandates — subject to the P/E leak.
- Persona/Big Five as vocabulary, and OCEAN as non-validation: Kocielnik (Big Five non-predictive), Song, PERSIST, Jiang/Peng/Yan (single-digit-pp human effects vs 80-pp targets).
- Stateless allocation dominated by the stated risk label: Ross & Lo — the mechanism behind the reframed claim.
- Transfer to medical triage: already executed (Menon §4.3).
- The M1–M4 framework: Arghal et al. (ICML 2026) is the behavioural+representational precedent; Arditi/Chen supply the direction recipes; Li supplies the attention measure and the split-softmax lever.

8.3 What remains unclaimed in the corpus

- The specific MAS/CI(MDD)/RG triple scored against a generating function, the three scripted regimes with hidden V_t, the T-calibration, the LCR construct, the $k \in \{1,5,25,100,\infty\}$ sweep as run (though Li's p-sweep is the same measurement).
- The 17/18 vs 16/18 split itself: content of a re-injected directive varied against OPPOSED allocation targets in a single behavioural readout in an identical market. Nearest analogues: Kocielnik's loss-averse vs gain-seeking framing (single session, no placebo, no trading); ContextEcho's V0/V2 content ablation (different surfaces, not opposed targets); Arike's emission-min vs profit-max goals (system prompt, not re-injected).
- Any 4.4×-style gap ratio: no paper measures a comparable statistic — and none supports a generic compounding law (Multi-IF front-loaded; SEQUOR post-reset; Dongre equilibrium).

8.4 Direct competitors and contradictions (with the number contradicted)

Paper	What it contradicts in FinPersona	FinPersona's number/claim at stake
Arike 2025 §5.1	Token distance from the system prompt is not what drives drift; in-context behavioural examples are.	"mandate has lost influence relative to the surrounding context" (§1); "market observations overwhelm initial instructions" (§3.2). Unsupported by design AND contradicted by nearest prior art.
Dongre 2025; Laban 2025; Multi-IF; SEQUOR	Multi-turn drift reaches bounded equilibria / is front-loaded / variance-dominated; no compounding law.	"grows monotonically from 1.0× to 4.4× by Q4" (§4.3.2), "compounds" (Conclusion).
Menon 2026 p.5	Frontier models show zero drift over 30 steps under adversarial pressure in a trading env.	Static agents "progressively abandon their target allocations" (cash 35.6% vs 49.4%) — cannot be the same phenomenon; the contradiction is with the causal story.
When Attention Closes pp.7–8, App. H	Accessibility decays 27–48% with recall at 100%; prepended periodic re-injection does not help.	Inference from behavioural drift to salience loss; "keeps the mandate active at every decision step" (§3.2).
Li et al. 2024 §4.2; Chen 2025 Fig. 4; Xiao 2024	Within-turn attention plateau; system-prompt content strongly encoded at last prompt token; attention to early tokens does not decay with distance.	Any attention-level "salience decay" in a fixed-length stateless prompt — M1 is flat by mechanism, not only by construction.
Kruthof 2026	Violation coexists with 97% recall; forgetting/decay framing rejected.	The name "Mandate Salience DECAY"; in a stateless design non-compliance is knows-but-violates by construction.
CLQT §7.11, p.19	HOLD rates of 44–50% were schema/tool-exhaustion artefacts; style drift ≈ 0 over 26 rounds.	MAS = 0.000 "perfect adherence" cells and the ISFJ cash shift — must be audited against parse/refusal/always-HOLD before being read as content effects (no such rates are reported).
ContextEcho §3.2 Q4	An imperative anchor lifts NO-DRIFT targets to ceiling — a compliance amplifier.	"memory ≫ placebo … consistent with MSD as a salience-based failure" (App. F) — equally consistent with amplification in the absence of decay.
Kocielnik App. I.9	Task-agnostic in-context text (Big Five items) shifts risk-taking (8.9→13.2 cards).	Reading the declarative placebo as a clean null — it is itself significant vs static ($p=0.041$), uniformly worse.
Song 2026 RQ3–4	Persona effects are lexical-cue-driven and do not transfer to realistic generation.	"framework-independent … behavioral content … drives the effects" (App. G) — the O3 result (vocabulary removed → penalty vanishes) fits Song better.
Jiang/Peng/Yan 2024 Table 7	Big Five explain ≈5% of equity-share variance; single-digit-pp effects; Neuroticism largest.	80-pp C_ideal spread; MBTI instrument omits Neuroticism.
AI-Trader; KTD-Fin; Hashimoto	Cash-holding/passivity is a spontaneous default for several models absent signals.	ISFJ 17/18 must be reported alongside the static cash baseline and the zero-trade share.

8.5 Implications for the M1–M4 programme (from the mech-interp corpus)

- M1: Li's attention decay is cross-turn with a within-turn plateau; Xiao shows attention to early tokens does not decay with distance; Chen shows the system prompt is strongly encoded at the last prompt token. In a stateless prompt M1 is flat by MECHANISM, not only by construction; any variation would track the observation O_t (market phase). Confirms the cut.
- M2: Arditi/Chen recipes transfer, but Chen's within-condition correlations (0.25–0.81), Arghal's linear-probe weakness and Templeton's few-shot-vector caveat predict a noisy, blunt single direction; defining the contrast by MAS risks circularity with the cash readout.
- M3: "ablate early, add back late" presupposes a trajectory that does not exist. The only version that speaks to "salience" is a MEDIATION test — ablate the direction (all layers/positions; Arghal's 0/456 single-layer result argues against single-layer patches) and ask whether the directive effect on the first action token vanishes; plus Li's split-softmax as an attention-boost lever. When Attention Closes App. D.5 is a risk signal (n=30, one model, one task), not a settled null. Confirms: post-P2, pre-registered, appendix at most.
- M4: ROME supplies causal tracing, not DLA (revision plan mis-cites it); the attribution tool in the corpus is Templeton-style gradient×feature; the mini-model reversal is mostly on closed models (Qwen2.5-7B the only open one). Confirms the cut.
- Naming: given Kruthof, Li and Chen, "salience decay" is unsupported for a stateless agent; "directive weighting under competing market signals" / knows-but-violates is what the corpus supports.

8.6 Corrections to Cycle 1's literature attributions (found by the full read)

13. "Kim/Li COLM 2024" is Li, Liu, Bashkansky, Bau, Viégas, Pfister & Wattenberg (COLM 2024, arXiv 2402.10962); the file is misnamed. Substance (SPR p-sweep, MMLU axis, baseline status) is correct.
14. DriftBench (Kruthof) does NOT show an "opposite capacity ordering" — within OpenAI and Google the larger variant drifts less (same direction as FinPersona's minis-worse); Sonnet 4.6 is the outlier. Correct statement: capacity ordering is provider-specific and does not transfer.
15. ContextEcho does have a behavioural task (judge-free S2 regex compliance; SWE-style argument-fidelity) — "no behavioural task and no ground truth" should be narrowed to "no long-horizon task scored against a generating function". It ALSO varies anchor content (App. G), so "ContextEcho varies deployment mode" is incomplete.
16. Meta (Wu et al.): always-on injection never falls below the no-memory baseline in Table 2; selective wins on macro average only; what hurts is ungrounded injection-only guidance and an untrained memory agent. "Near-verbatim" overstates the definitional match (same concept, different words).
17. When Attention Closes App. D.5 is not "well-powered": n=30 pairs, Mistral-7B, fact recall, one layer block; the authors call it "descriptive localization, not causal identification". App. H is an "analogous" prior 3-arm design, not "identical": prepended, recall task, forced sliding-window mask, qualitative, null for both arms.
18. Jiang/Peng/Yan: "5–8 pp equity swing" is derived (4.8–8.7 pp for Neuroticism; 2.3–4.7 for Openness), not reported; Neuroticism is largest for total/retirement equity only; the paper's headline is "Neuroticism and Openness".
19. Arike's horizon is up to 74 steps / >100K tokens (settings 3–4), not "≈30 steps"; Arike's token-distance ablation is a context filler, not a placebo for an appended directive (C11 over-credits it).
20. C10 omitted direct anticipations: Mehri 2026 App. E and Dongre 2025 (per-turn/sparse reminders with model-specific reversals); the Always-On survey §6.4.3 is the strongest general prior.

21. C1 omitted Dongre 2025 (the closest dynamical formalisation in the corpus); C2 omitted Song 2026 and should not cite Arike (whose Finding 9 shows stated goals TRACK behaviour).
22. SEQUOR's manifest figures (−11/−40/−9%) are the abstract's best-model numbers; averages are −26/−38/−27% (and −23% for the replacement regime proper).
23. Everitt "supports MSD ≠ capability" over-reaches: Everitt's App. E shows degradation NOT caused by context length — it cuts against the mechanism.
24. Not raised by Cycle 1: the paper mis-cites Hong 2025 (retrieval, not instruction following) in its intro; the revision plan mis-cites ROME for direct logit attribution; Table 2 lists MACD/dividend yield that are not in the prompt.

Cycle 1 attributions verified CORRECT by the full read: CLQT §3.5 trigger / a_target undefined / investment-target not ablated alone / §8.8 future work; CLQT coherence gap and inter-axis |r| = 0.23; Ross & Lo 57–88% and input-sensitivity criterion; Kruthof KBV 8–99%; ContextEcho's six facts; Menon ER triage; Kocielnik's Big Five non-prediction; Machine Spirits, Multi-IF, PushBench, Arghal, Sawant, Drift-Bench-ND manifest descriptors; the ABIDES→Hashimoto correction; "Meta built it" (C18).

8.7 Mandatory citation set and differentiation

Sixteen papers the revision must cite (none currently cited), each with the differentiation the authors need — full sentences are in the per-paper files:

Paper	One-line differentiation
Arike et al. 2025	They measure goal drift of tool-using LLM traders under adversarial pressure in a history-carrying loop and find drift is pattern-matching, not token distance; we study stateless single-step agents and isolate content of a recency-appended directive against a placebo.
Menon et al. 2026	Frontier agents hold a system-prompt trading goal for 30 steps under pressure; user-turn instructions override system goals for some families — we hold context fixed and vary only the content of a user-turn directive.
Li et al. 2024 (COLM)	They show cross-turn attention decay and SPR as a p-swept baseline with an MMLU cost; our agents are stateless, so we claim no cross-turn decay and use per-step injection as a content probe.
ContextEcho 2026	Their user-turn anchor with a length-matched filler control acts as a generic compliance amplifier across 23 models; the amplifier reading is the null we must reject before claiming anything beyond content-modulated amplification.
Ross & Lo 2026	Stateless allocations are dominated by the stated risk label; we report the directive's share of predictive weight alongside observation features rather than treating MAS shifts as decay.
Kruthof 2026 (DriftBench)	Models violate constraints they can restate; since our agents see the mandate on every call, violation is a weighting failure by construction, and we name it so.
Dongre et al. 2025 (Drift No More)	Multi-turn drift is a bounded process; our quartile widening in the crash reflects regime phase, not accumulated context; we claim no unbounded growth.
When Attention Closes 2026	Naive prepended periodic re-injection did not restore recall under forced attention closure; ours is appended, on an allocation outcome, in stateless agents — not a rerun of their null.
Wu et al. 2026 (Meta)	They inject selected task-state reminders and find selective beats always-on on balance; we inject a fixed behavioural directive and find its effect's sign is set by content.
CLQT 2026	Diagnoses a stating-vs-doing gap under one fixed mandate with a threshold-triggered drift warning and names mandate variation as future work; we vary the mandate itself.
Kocielnik et al. 2026	Persona prompts move self-reports not behaviour, while in-context policy framing moves risk-taking; we measure allocations over 200 decisions and contrast a directive with declarative boilerplate.
Mehri et al. 2026	Per-turn goal re-injection improves alignment with model- and dimension-specific reversals; we observe the analogous bidirectionality for financial mandates with a placebo.
Hashimoto et al. 2025	LLM buy/sell intentions inside a multi-agent LOB market; we isolate a single agent against an exogenous path with an exact hidden value.
Song et al. 2026	Persona-prompt effects are lexical-cue-driven; our numeric-only ablation removing the ENTJ penalty is consistent with their account.
Jiang, Peng & Yan 2024	Big Five explain ≈5% of equity-share variance; our targets are stipulated, not derived.

Laban 2025 / Always-On survey 2026	SNOWBALL repetition and the baseline-beats-memory literature; our arms are episodic by design and isolate one prompt element.

Also recommended: PERSIST, AI-Trader, Machine Spirits, SEQUOR, Multi-IF, Everitt, Fin-Bias, KTD-Fin, Abdelnabi, Arditi/Chen/Arghal (if any M-series survives), Xiao (if M1 ever runs). Reword the Hong and Liu citations. Do not cite AlphaSAGE or Drift-Bench-ND (footnote the name collision).

8.8 Consequences for the verdict

The full read does not change the GO/NO-GO but sharpens its conditions and lowers the novelty ceiling: (i) the contribution must be claimed as novel in domain and readout, not in injection logic; (ii) ContextEcho's compliance-amplifier reading becomes the explicit null hypothesis for the memory arm — P2's wrapper-only and directive-placebo arms are its test; (iii) Ross & Lo's heuristic-collapse result must be cited as the reason an opposed-target split is expected; (iv) Kruthof/Li/Chen settle the naming — "salience decay" goes; (v) CLQT/AI-Trader/KTD-Fin make a parse/refusal/always-HOLD audit and a static-cash-baseline report mandatory before any cash-shift claim; (vi) the 16-paper citation set with differentiation is non-negotiable, with Li et al. correctly attributed.

9. GO / NO-GO recommendation

GO — CONDITIONALLY (unanimous across all three reviewers and the orchestrator), on the reframed programme with P0→P2 as the critical path.

Binding conditions (merged; violations convert the recommendation to NO-GO):

25. No new compute before P0 lands. The rewrite of the central claims — stateless architecture, mandate-absent baseline, 4.4× withdrawal, crash downgrade, Figure 3/4/5 corrections — precedes every experiment; otherwise new data are poured into a broken frame.
26. No mechanistic-interpretability spend this cycle. M1 and M4 are cut; M2/M3 exist at most as a post-P2, pre-registered, null-committed appendix.
27. The pooled crash claim is withdrawn or explicitly labelled directional-but-not-significant (model-level $p \approx 0.10$–0.24). The ISFJ-scoped claim ($p=0.0038$) may stand only with the zero-trade mechanism disclosed. Any retained crash analysis must discuss the δ-sweep's near-vacuity.
28. P2 (swapped mandate, directive placebo, wrapper-only, dose-response) is pre-registered with predictions stated in the paper, and runs before resubmission. Without it the surviving contribution is exposed to the one objection — a same-signed cash shift refracted through opposed targets — that this cycle could not dismiss from existing data.
29. All classifier-derived numbers are quarantined until retrained with documented, non-circular provenance; App. H's operational recommendations are removed in any case.
30. One metrics pipeline: a single implementation of MAS/CI(MDD)/RG feeding every table and figure, making a four-way cohort split for one cell impossible by construction; released with the repo.
31. (From the full-corpus read) Degeneracy audit before any cash-shift claim: parse-failure, refusal and always-HOLD/zero-trade rates per cell, and the static-arm cash baseline, reported alongside every memory-vs-static contrast (CLQT §7.11; AI-Trader; KTD-Fin).
32. (From the full-corpus read) Novelty claimed as domain and readout only; ContextEcho's compliance-amplifier finding stated as the null hypothesis and tested by P2's wrapper-only/directive-placebo arms; Ross & Lo cited as the reason the opposed-target split is expected; the construct renamed away from "salience decay" (Kruthof/Li/Chen).

33. (From the full-corpus read) The 16-paper mandatory citation set (Section 8.7) with differentiation sentences, Li et al. (COLM 2024) correctly attributed, Hong/Liu citations reworded, ROME not cited for DLA.

Reject-package tripwires (any of these in a resubmission and the recommendation is NO-GO regardless of added work): retaining MSD-as-decay or "as market context accumulates"; the 4.4×/monotone-compounding claim; "re-grounding" vocabulary for a mandate-absent baseline; pooled-N=270 inference; unstated-target MAS without the communicated-target repair; classifier-based operational guidance; an injection-frequency ablation that does not cite Li/Kim COLM 2024; or adding M1–M4 on top of the current claims.

What is genuinely worth saving, and why GO rather than NO-GO: the 17/18-vs-16/18 content-modulated pattern is real, exactly reproducible, model-level significant, replicated under OCEAN vocabulary and at T=800, and unclaimed in a 47-paper corpus; the paired fully-scripted grid is cheap to extend and unusually reproducible (the tables reproduce digit-for-digit — the rot is in figures and framing, not data); and the decisive discriminating experiment costs on the order of a hundred cheap simulations. The paper that survives this review is smaller than the paper that was submitted, but it is real.